



Zoo 3.0 — Full Post-Quantum Stack

Every signature, every commitment,
every proof in PQ-safe schemes

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Abstract

Zoo 3.0 is the full post-quantum era of the Zoo stack. Zoo 2.0 (October 2024) shipped a post-quantum L2 EVM on LUX with a precompile suite covering NIST PQC standards [10]; Zoo 3.0 (2025) hardens *every* hot-path primitive across the Zoo ecosystem to PQ-safe schemes. Specifically: every consensus signature is a triple cert (BLS₁₂₋₃₈₁ classical fast path, RINGTAIL Ring-LWE threshold proof, ML-DSA-65 identity proof) inherited from the LUX QUASAR 3.0 lane (LP-020); every governance commitment is a ML-DSA-65-signed Merkle root; every loan agreement under the Zoo NFT Liquidity Protocol is RINGTAIL-signed and bridge-attested; every inference receipt and training-data attribution carries PQ-safe witnesses. The Zoo Bridge committee re-keys to threshold ML-DSA + RINGTAIL on M-Chain. The Zoo DAO migrates governance keys to threshold ML-DSA-65. zLLM commitments for training artefacts, governance votes, and inference receipts adopt PQ-safe Merkle accumulators. This paper documents the full PQ stack, the migration roadmap, and the cross-references to LUX LP-020 (Quasar 3.0), LP-070 (ML-DSA), LP-073 (RINGTAIL), and LP-075 (BLS). Spec freeze: 2025-10-31. Phased rollout through 2025; full activation aligned with LUX QUASAR 3.0 cutover.

Keywords: Zoo 3.0, post-quantum, RINGTAIL, ML-DSA-65, BLS₁₂₋₃₈₁, triple cert, QUASAR 3.0, zLLM, NFT Liquidity Protocol

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1 Context: from 2.0 to 3.0

ZOO 3.0 is the post-quantum hardening epoch of the ZOO stack. This section lays out where ZOO stood at the start of 2025, what ZOO 2.0 left unfinished, and what 3.0 commits to deliver.

1.1 The arc up to 3.0

1.0 (2021, BSC). The original ZOO whitepaper of October 31, 2021 [12] positioned ZOO as a wildlife-conservation platform on Binance Smart Chain. That paper coined the term “NFT Liquidity Protocol” — the mechanism by which NFTs become productive collateral inside a yield-bearing system. Cryptographic posture was inherited from BSC: ECDSA over secp256k1, Ethereum-style Merkle Patricia tries, no post-quantum guarantees.

2.0 (October 2024, Lux L2). ZOO 2.0 [10] re-launched on the LUX Network as an EVM-compatible L2. The 2.0 paper introduced 20 native precompiles including NIST PQC standards (ML-DSA, SLH-DSA, ML-KEM), threshold signatures (FROST, CGGMP21), privacy primitives (FHE, ZKP, ring signatures), and an AI Mining precompile. 2.0 delivered *first-pass* post-quantum: PQ primitives were available to applications via precompile, but the chain consensus, governance, bridge, and the legacy 2021 NFT Liquidity Protocol commitments still relied on classical cryptography for some edges.

3.0 (2025, full PQ). ZOO 3.0 closes those edges. The contract: every signature on the hot path, every commitment in governance, every proof in the bridge, every receipt under the NFT Liquidity Protocol uses post-quantum schemes. PQ is no longer optional or per-application; it is the chain-wide default.

4.0 (2026, GPU-native L1). ZOO 4.0 [11] graduates ZOO from L2 application to sovereign L1 with 100% GPU-powered consensus, EVM, DEX, AI mining, and FHE. 4.0 inherits 3.0’s full-PQ posture unchanged.

1.2 The threat model that drives 3.0

ZOO’s threat model in 3.0 is the harvest-now-decrypt-later adversary plus a near-term Q-day adversary. Harvest-now is already operational: a passive observer can snapshot any classical signature ever recorded on a public chain and decrypt years later when a cryptographically relevant quantum computer (CRQC) becomes available. This means classical signatures recorded today on hot-path data (governance commitments, bridge attestations, NFT loan agreements) already need to be PQ-safe, because the value those signatures authenticate may persist far past Q-day.

A near-term Q-day adversary further means the chain must be able to validate the integrity of pre-Q-day records using post-Q-day verifiers. ZOO 3.0’s triple-cert design (see §2) addresses this directly: every block carries a classical proof and two independent post-quantum proofs, so even when the classical layer is broken the post-quantum layers continue to validate.

1.3 What 3.0 changes (one-page summary)

Component	2.0 (October 2024)	3.0 (2025)
Consensus signatures	inherited L2 from LUX 2.x	triple cert via QUASAR 3.0 [4]

Governance keys	ECDSA per signer	threshold ML-DSA-65 + RINGTAIL
Bridge attestations	classical multisig	threshold ML-DSA on M-Chain (LP-016, LP-017)
zLLM commitments	SHA-256 Merkle	PQ-safe accumulator + ML-DSA-signed root
NFT Liquidity loans	ECDSA loan agreements	RINGTAIL-signed loan agreements
DAO vote receipts	on-chain ECDSA	ML-DSA-65 signed, optional FHE ballot
Inference receipts	A-Chain TEE quote	TEE quote + RINGTAIL-anchored attribution
Wallet signing keys	secp256k1 only	dual-key (secp256k1 + ML-DSA-65)

The remaining sections expand each row into its specification and migration plan.

1.4 What 3.0 explicitly does *not* change

- ZOO stays an L2 on LUX; the L1 graduation is a 4.0 concern [11].
- EVM bytecode semantics are unchanged; only the signature scheme over transactions and state commitments moves to PQ-safe forms.
- Application contracts written for 2.0 continue to work in 3.0; new precompiles for RINGTAIL verification are additive, not breaking.
- Token standards (ZRC-20, ZRC-721, ZRC-1155) are unchanged; only the underlying signing key types are extended.

1.5 Why 2025 is the right epoch for full PQ

NIST finalised FIPS 204 (ML-DSA) in August 2024 [14], which gives ZOO a standardised lattice signature with broad ecosystem support. The LUX Network shipped QUASAR 3.0 triple consensus [4] in 2025, giving ZOO a chain-level substrate for triple-cert proofs without per-app integration. The remaining work in 2025 is the application-layer migration: governance, bridge, NFT Liquidity, zLLM commitments, wallet keys. ZOO 3.0 sequences that migration through the rest of the year so that 4.0 inherits a uniformly post-quantum stack.

2 Triple-Cert Adoption

ZOO 3.0 inherits the LUX QUASAR 3.0 triple-cert lane defined by LP-020 [4], with each cert layer specified in its own LP: LP-075 for BLS₁₂₋₃₈₁ [7], LP-073 for RINGTAIL [6], LP-070 for ML-DSA-65 [5].

2.1 The three lanes

A QUASAR 3.0 cert is the conjunction of three independent proofs:

1. **BLS₁₂₋₃₈₁ aggregate (LP-075)**. Classical fast-path proof. 48 bytes. Pairing-based. Falsifies under a CRQC; remains useful as the low-latency check that gates pre-final block propagation.
2. **RINGTAIL threshold (LP-073)**. Post-quantum threshold proof built on Ring-LWE. Threshold t -of- n aggregation across the validator set. Approximately 2.0–2.4 KB depending on threshold and committee size.
3. **ML-DSA-65 identity (LP-070)**. Per-validator post-quantum identity proof, FIPS 204 conforming. Approximately 3.3 KB. Establishes which validator authorised a given cert layer.

The triple-cert is the conjunction of all three. A block with only BLS is invalid in 3.0 even if a majority of stake signed it classically.

2.2 Why triple, not double

A double cert (classical + one PQ) leaves a single-point failure on the post-quantum side. If the chosen lattice scheme has a structural flaw that downgrades its security, the chain has no second PQ proof to fall back on. QUASAR 3.0’s design intent is to require two independent PQ assumptions: RINGTAIL relies on Ring-LWE hardness; ML-DSA relies on the hardness of M-LWE and M-SIS. A simultaneous break of both — on different lattice problems with different parameter regimes — is materially less likely than either individually. The classical BLS layer remains for fast-path liveness while the chain is healthy, and as a defence-in-depth witness while a CRQC is plausible but not yet operational.

2.3 What 3.0 inherits unchanged

ZOO 3.0 does not modify QUASAR 3.0 itself; ZOO consumes it as a chain-level service. The pieces ZOO inherits from LP-020:

- **TripleSignRound1**. The kernel that orchestrates parallel execution of the three cert lanes within a single round.
- **Photon committee selection (LP-105)**. VRF-driven selection of the round’s committee, deterministic given the previous QUASARCert.
- **Wave threshold voting (LP-105)**. Threshold accumulation of signatures inside each lane.
- **Focus confidence accumulation (LP-105)**. The check that all three lanes have crossed their respective thresholds before the cert is finalised.

2.4 Where Zoo plugs in

Zoo’s L2 chain settles state commitments to the L1 QUASARCert. In 3.0, those commitments are themselves signed under the triple-cert discipline, so a ZOO block is final when:

1. the ZOO sequencer signs the block with BLS, and
2. the ZOO sequencer submits the block hash to LUX QUASAR 3.0 for inclusion, and
3. a quorum of LUX validators emit a QUASARCert that anchors the ZOO block.

The L2 inherits the L1’s PQ guarantees through the cert. Pre-3.0, ZOO L2 also signed the inclusion proof with classical-only multisig at the operator boundary; 3.0 adds the requirement that the operator multisig be threshold ML-DSA-65 rather than ECDSA.

2.5 Verifier interface

A ZOO 3.0 light client must verify three proofs per epoch:

```
1 struct QuasarCert3 {
2     bls::Aggregate      bls_proof;           // 48 bytes
3     ringtail::ThresholdSig ringtail_proof; // ~2.0-2.4 KB
4     mldsa::Aggregate    mldsa_proof;       // ~3.3 KB * t
5     Hash                block_root;
6 };
7
8 bool verify_3_0(const QuasarCert3& c, const ValidatorSet& vs) {
9     if (!bls_verify(c.bls_proof, c.block_root, vs)) return false;
10    if (!ringtail_verify(c.ringtail_proof, c.block_root, vs)) return false;
11    ;
12    if (!mldsa_verify(c.mldsa_proof, c.block_root, vs)) return false;
13    return true;
14 }
```

Listing 1: Triple-cert verifier sketch

A failure on any lane fails the cert. There is no “best of” fallback.

2.6 Performance budget

The dominant cost in cert verification under 3.0 is the ML-DSA-65 aggregate, because it scales linearly in the number of distinct signers in the threshold whereas BLS aggregates to a single 48-byte object. Empirically, on the LUX validator hardware class targeted for 3.0, the verifier closes at well below the per-block latency budget ([8], [4]). ZOO’s client side picks up the cost as a one-time verification per finalised LUX epoch; per-transaction cost is unchanged.

3 zLLM Post-Quantum Commitments

zLLM is ZOO’s training-free experience-library adaptation system: a base model is frozen, and an experience library curated by the DAO is injected at inference. The system needs commitments at three points: the experience-library state, the governance vote that admitted each experience, and the inference receipts that record which experiences were used.

In 3.0 each of these three commitments uses post-quantum primitives.

3.1 Experience library commitment

The experience library is a Merkle accumulator over individual experience entries (a tuple of prompt context, expected behaviour, provenance, and metadata). Pre-3.0 the accumulator used SHA-256 over classical Merkle paths and was signed by the curating multisig with ECDSA. In 3.0:

- Hash function: SHA3-256 (Keccak-f) for collision resistance under Grover-bounded adversaries; the 128-bit post-quantum security target requires the underlying hash to provide 256-bit classical resistance.
- Accumulator: a sparse Merkle tree (still classical structure; PQ resistance comes from the hash and the signing key).

- Signing key on the accumulator root: threshold ML-DSA-65 over the curators set, t -of- n where $t \geq \lceil 2n/3 \rceil$.

The experience library commit transaction posts the new root and the threshold ML-DSA signature. Verifiers re-derive the threshold public key from the on-chain curators set and check the signature.

3.2 Governance vote commitments

zLLM experience admission is a DAO vote. Each ballot is a ML-DSA-65 signature over the experience hash, the voter’s weight, and the vote direction. Tally is a deterministic Merkle root over the ballots, signed by the tally proposer with ML-DSA. Optional FHE mode (per [13]) keeps ballot direction private until the tally is opened, but the public result is still authenticated by ML-DSA.

A vote receipt for an admitted experience contains:

1. The experience hash.
2. The governance proposal hash.
3. The tally Merkle root.
4. The ML-DSA signature on the tally root.
5. The block height of the tally on the L2.
6. The reference QUASARCert from LUX L1.

A verifier can reconstruct the entire admission chain from these six fields.

3.3 Inference receipts

When a zLLM inference uses the experience library, the inference receipt records which experiences were retrieved. The receipt schema in 3.0:

```

1 struct InferenceReceipt3 {
2     Hash      base_model_id;           // pinned weights digest
3     Hash      experience_root;         // accumulator root at retrieval
4     Hash []    experience_paths;       // Merkle paths for retrieved
        entries
5     Hash      prompt_digest;           // hash of input (or FHE handle)
6     Hash      output_digest;          // hash of output (or FHE handle)
7     TeeQuote   tee_quote;              // attestation of execution env
8     MlDsa::Sig signing_proof;          // worker's ML-DSA-65 signature
9     Hash      a_chain_anchor;          // commitment to A-Chain (LP-134)
10 };

```

Listing 2: 3.0 inference receipt schema

The receipt establishes:

- which model produced the output (`base_model_id`),
- which experiences shaped it (`experience_paths`),

- that the execution happened inside a TEE-attested container (`tee_quote`),
- that the worker authenticated under a post-quantum signing key (`signing_proof`),
- that the receipt is anchored to A-Chain (LP-134) for cross-chain verifiability.

3.4 Attribution graph

zLLM rewards flow back to upstream contributors: experience curators, training-data donors, evaluators, and reviewers. The attribution graph is itself a series of commitments. In 3.0 every node in the attribution graph is signed with the contributor’s ML-DSA key, the edge between contributor and experience is a Merkle inclusion proof under the experience library root, and the disbursement transaction is a triple-cert-anchored LUX L1 transaction [4].

This means a long-tail contributor whose data shaped a Zen model in 2024 still receives correctly-attributed disbursements years later under post-quantum verification — the historical record is not invalidated by Q-day.

3.5 Migration of pre-3.0 commitments

Experience-library entries created before 3.0 carry classical commitments. The migration plan:

1. **Re-sign campaign.** Curators re-sign the legacy accumulator root with the new threshold ML-DSA-65 key. The on-chain commitment now carries both the legacy ECDSA and the new ML-DSA signatures.
2. **Sunset window.** For 90 days post-3.0 spec freeze, verifiers accept either signature.
3. **PQ-only enforcement.** After the sunset window, only ML-DSA signatures are accepted. Legacy ECDSA-only commitments must be re-signed before that date or be considered un-attestable.

This is the same pattern adopted by the DAO key migration (§6).

4 Zoo Bridge Post-Quantum Attestations

The ZOO Bridge moves assets between ZOO L2 and external chains: LUX C-Chain (canonical), Ethereum, BSC, and other EVM ecosystems for which the bridge committee has stood up adapters. In 2.0 the bridge attestation was a classical multisig over the mint event. In 3.0 the bridge attestation lives on M-Chain and is secured by post-quantum threshold signatures per LP-016 [2] and LP-017 [3].

4.1 Bridge attestation flow under 3.0

1. **Source observation.** The bridge committee observes a finalised burn or lock event on the source chain.
2. **Quorum vote.** The committee runs a threshold ML-DSA-65 ceremony on M-Chain: each member submits an ML-DSA signature over the event digest. The threshold is t -of- n with $t \geq \lceil 2n/3 \rceil$.
3. **RINGTAIL co-attestation.** The committee additionally emits a RINGTAIL threshold proof over the same digest. Both proofs must succeed for a mint message to be valid.

4. **B-Chain emission.** The pair of proofs is wrapped into a B-Chain bridge message (LP-016) and delivered across to the destination chain.
5. **Destination mint.** The destination contract verifies both proofs — ML-DSA via on-chain pre-compile and RINGTAIL via the verifier shipped in 3.0 — and only then mints.

4.2 Why two PQ proofs on the bridge

Bridges are the highest-value targets in the ecosystem. A break of either lattice scheme would compromise mint authority on the destination chain. Requiring both RINGTAIL (Ring-LWE) and ML-DSA (Module-LWE/SIS) means the attacker must break two distinct lattice problems — not just two parameter sets of the same family — to forge a mint. Combined with the source-chain finality requirement and the threshold $t \geq \lceil 2n/3 \rceil$ on the committee, this is the strongest practical defence against both Q-day and committee compromise.

The classical BLS proof is also recorded for fast-path verification on legacy explorers and as defence-in-depth, but it is not load-bearing in 3.0; verifiers ignore it for mint authorisation.

4.3 Committee key schedule (LP-017)

Per LP-017 [3], the committee re-keys on a defined schedule:

- Every 90 days the committee runs a key-refresh ceremony, rotating each member’s ML-DSA and RINGTAIL share.
- The ceremony is itself M-Chain-anchored: the new shares are committed under the previous round’s threshold key, so an attacker who compromises a single member at one round cannot persist the compromise through the rotation.
- Members who fail to participate in two consecutive rotations are removed from the committee by L1 governance.

4.4 Cross-chain message format

A 3.0 bridge mint message carries:

```

1 struct BridgeMintMsg3 {
2     ChainId    src_chain;
3     ChainId    dst_chain;
4     Hash       src_event;           // finalised burn/lock event
5     Hash       asset;
6     Address    beneficiary;
7     Amount     amount;
8     Hash       committee_epoch;    // identifies key schedule epoch
9     MlDsa::Sig  mldsa_threshold;   // t-of-n ML-DSA-65
10    Ringtail::Sig ringtail_threshold; // t-of-n Ringtail
11    Bls::Agg     bls_legacy;        // for fast-path observers, advisory
12 };

```

Listing 3: Bridge mint message v3

The destination-chain verifier rejects on a missing or invalid ML-DSA or RINGTAIL proof. The BLS field is optional under 3.0.

4.5 2021 BSC migration adapter

The ZOO Bridge supports a one-way migration adapter for legacy 2021-cohort BSC ZOO holders. Holders burn the legacy BEP-20 token; the bridge committee attests the burn under the 3.0 attestation discipline; the holder receives the ZOO 3.0 native asset on ZOO L2. The legacy-side burn cannot itself be PQ-attested (BSC does not implement ML-DSA or RINGTAIL verification), but the destination-side mint is fully PQ-attested. This is the correct posture: the chain we control is post-quantum; the chain we are migrating from continues to be classical for as long as upstream chooses.

4.6 Failure modes and recovery

- **Quorum failure.** If the committee cannot reach the threshold, the message is not emitted. Mint does not happen. No silent fallback to a smaller quorum.
- **Single-member compromise.** The threshold structure absorbs up to $n - t$ compromised members. Compromise above that triggers L1 governance to slash and rotate.
- **Lattice scheme break.** If a published break of either RINGTAIL or ML-DSA surfaces, the bridge halts pending emergency re-parameterisation; the committee proposes new parameters via L1 governance and re-keys before resuming.

5 NFT Liquidity Protocol Post-Quantum

The NFT Liquidity Protocol — the term ZOO coined in the October 31, 2021 whitepaper [12] — turns NFTs into productive collateral inside a yield-bearing system. In 3.0 the loan, repayment, and liquidation primitives are all post-quantum authenticated.

5.1 Why this matters specifically for NFT collateral

An NFT loan is a long-lived bilateral commitment: the borrower locks the NFT, the lender extends a loan, and the agreement persists for the loan duration (months to years). A classical signature on that agreement is exactly the kind of long-lived hot-path credential that harvest-now-decrypt-later most threatens. If the agreement expires post-Q-day, the holder of the original loan agreement may discover their authority has been retroactively forged. 3.0 closes that gap.

5.2 Loan agreement schema (3.0)

```
1 struct NftLoanAgreement3 {
2     Hash      nft_id;                // ZRC-721 token reference
3     Address   borrower;
4     Address   lender;
5     Amount    principal;
6     Rate      apr;
7     Duration  term;
8     Hash      collateral_appraisal;  // F-Chain FHE handle (optional)
9     MlDsa::Sig borrower_sig;        // ML-DSA-65 signature
10    Ringtail::Sig lender_sig;        // Ringtail signature, threshold
11    Hash      l1_anchor;             // Quasar 3.0 cert reference
12 };
```

Listing 4: 3.0 NFT loan agreement schema

The borrower signs with ML-DSA-65 to bind individual identity. The lender signs with RINGTAIL, threshold-capable so that pooled-lender markets (lender DAOs, lending pools) can present a single threshold signature standing in for many participants. Both signatures are required for the loan to bind.

5.3 Confidential appraisal

The optional `collateral_appraisal` field is a handle to an F-Chain FHE ciphertext holding the appraised value of the NFT. A confidential market can run liquidation logic over the ciphertext without exposing the appraised value to public observers. This is useful for high-value collections where revealing appraisal is itself information leakage. The 3.0 FHE substrate is the same one used elsewhere in the stack (LP-013, Confidential ERC-20 LP-067), so appraisals can be composed with confidential balance transfers without breaking PQ guarantees.

5.4 Liquidation flow

Liquidation is the highest-stakes path in the protocol because it transfers the NFT away from the borrower. The 3.0 liquidation flow:

1. Liquidation oracle (an A-Chain attested process) reads the appraisal-vs-debt threshold.
2. Oracle issues a liquidation event signed under threshold ML-DSA.
3. Loan contract verifies the oracle’s threshold signature against the on-chain oracle set’s public key.
4. Contract transfers NFT to lender (or auctions to the lending pool, depending on agreement type).
5. Liquidation receipt is anchored to LUX QUASAR 3.0 cert [4], providing a triple-cert-final record of the transfer.

A classical-only oracle compromise (a future Q-day adversary forging the oracle’s classical signature) cannot trigger liquidation in 3.0 because the liquidation contract does not honour classical-only signatures.

5.5 Repayment flow

Repayment closes the loan and unlocks the NFT. The borrower submits a repayment transaction signed with ML-DSA; the contract verifies the signature, transfers the principal plus accrued interest to the lender, and releases the NFT from escrow. The repayment receipt is anchored to L1 cert and is independently verifiable post-Q-day.

5.6 Pooled lending

Pooled lender markets aggregate many lenders behind a single RINGTAIL threshold key. The threshold structure means individual lender shares can be added or removed without re-issuing every outstanding loan: the pool’s threshold public key is stable across lender membership changes inside the agreed parameter set. When the pool needs to rotate its lattice parameters (a security-driven upgrade), it emits a one-time PQ-signed parameter-update event, and all in-flight loans pin to the parameter epoch in which they were signed.

5.7 Migration of pre-3.0 loan agreements

1. All open 2.x loan agreements are walked by an automated migration job.
2. For each open loan, both counterparties are notified to co-sign a 3.0 loan agreement that supersedes the 2.x form, carrying the same economic terms.
3. Loans whose counterparties do not co-sign within the 90-day sunset window are flagged “classical-only” on the UI; liquidation against these loans requires an additional consensus check.
4. Newly-issued loans after the 3.0 cutover are PQ-only.

This is the same sunset pattern used for governance keys (§6) and zLLM commitments (§3).

6 DAO Key Migration

The Zoo DAO authorises treasury disbursements, governance votes, contract upgrades, and per-LLM chain admissions. In 2.x the DAO’s authority surface used classical multisig and ECDSA. In 3.0 the entire DAO authority moves to threshold post-quantum signing.

6.1 What moves

- **Treasury custody key.** Threshold ML-DSA-65 on M-Chain (per LP-017 schedule [3]).
- **Governance proposal-signing key.** Threshold RINGTAIL for proposal hashes (PQ-safe with smaller per-signer state than ML-DSA, useful for high-frequency proposal flows).
- **Contract-upgrade authority key.** Threshold ML-DSA-65 with optional human-in-the-loop second factor (a hardware-key-attested ML-DSA from each upgrade-authorised member).
- **Per-LLM chain admission key.** Per-chain threshold ML-DSA, scoped so that a compromise of one chain’s admission key does not propagate to others.

6.2 Why threshold ML-DSA + Ringtail and not just one

The DAO runs two independent PQ schemes for the same reason as the QUASAR 3.0 cert design (§2): two different lattice problems, so a structural break in one does not unlock the other. Treasury custody — the highest-stakes authority — uses ML-DSA-65 because FIPS 204 conformance maximises ecosystem auditability. Proposal-signing uses RINGTAIL because its smaller per-signer state suits high-frequency vote production.

6.3 Ceremony for the migration

The DAO cannot simply self-rotate to PQ keys without a public ceremony, because the legacy ECDSA holders need an auditable transfer event. The ceremony:

1. **Pre-announcement.** A governance proposal, signed under the legacy multisig, announces the migration window and target threshold parameters.
2. **Distributed key generation (DKG).** Each member runs a ML-DSA DKG to produce the new threshold key shares. Output is the threshold public key plus per-member secret shares.

3. **Co-signed handover.** A single transaction signed by *both* the legacy ECDSA multisig and the new threshold ML-DSA key transfers custody from the legacy address to the threshold address.
4. **Sunset.** The legacy multisig is retired after a defined window during which any member can challenge the ceremony with a fraud proof.

A failed ceremony reverts to the legacy multisig and the migration proposal must be re-issued. The ceremony is itself anchored to QUASAR 3.0 cert [4], providing triple-cert finality on the handover record.

6.4 Vote-signing migration

Voter wallets in 3.0 carry dual keys: a legacy ECDSA key for backwards compatibility, and a new ML-DSA-65 key. During the sunset period both signature types count toward quorum. After sunset, only ML-DSA ballots count; legacy-only voters who have not co-signed a wallet upgrade are silently dropped from the eligible set.

The wallet’s UX presents the dual-key state explicitly so users understand which signature scheme is in use for each action.

6.5 Optional FHE ballots

Per the holographic-consensus DAO design [13], ballots can be cast in FHE-encrypted form so vote direction is private until tally. In 3.0 the encryption is layered with the PQ signature: the ballot is FHE-encrypted, then the FHE ciphertext is signed under the voter’s ML-DSA key. The tally process operates over ciphertexts; the result is decrypted only at the published tally round. Voter authentication remains PQ-safe even though vote content is privacy-preserving.

6.6 Recovery in case of share loss

Threshold key share loss is a known operational hazard. The 3.0 recovery procedure:

- If $\leq n - t$ shares are lost, the threshold remains operable. Lost members are removed via governance and the committee re-keys at the next rotation epoch.
- If more than $n - t$ shares are lost, the threshold is broken; recovery requires a governance-authorized re-DKG signed by a quorum of the remaining live members plus a delay-locked failsafe authority that can authorise re-key after a multi-week timelock if even the remaining live members are insufficient.
- All recovery actions are publicly logged and anchored to L1 cert.

6.7 Effect on per-LLM chain governance

Per-LLM chains [13] run their own local governance for model-specific decisions (parameter votes, experience-library admissions, dataset whitelist changes). Each per-LLM chain inherits the 3.0 governance template: dual-key ballots, threshold ML-DSA for binding decisions, optional FHE voting. This means a chain compromise on one LLM chain does not require a global Zoo DAO re-key.

7 Activation Roadmap

ZOO 3.0 is a phased migration through 2025, ending with a fully post-quantum stack ready to be inherited by ZOO 4.0 [11] on its February 14, 2026 activation.

7.1 Phase 0 — LP coordination (2025-Q1 to 2025-Q2)

The LUX LPs that 3.0 depends on land in this window:

- LP-070 (ML-DSA-65 identity lane) [5].
- LP-073 (RINGTAIL threshold signature) [6].
- LP-075 (BLS₁₂₋₃₈₁ aggregate lane) [7].
- LP-020 (QUASAR 3.0 triple cert composer) [4].
- LP-016 (B-Chain bridge VM) [2].
- LP-017 (threshold bridge committee key schedule) [3].

Zoo’s role in Phase 0 is to participate in the LP review process and ensure that the ZOO L2 settlement contract, the bridge committee adapter, and the DAO key custody specifications can compose cleanly with each LP. No on-chain action on Zoo’s side.

7.2 Phase 1 — bridge re-keying (2025-Q3)

The bridge committee runs the first 3.0-discipline DKG ceremony (§4). The committee’s ML-DSA + RINGTAIL threshold key replaces the classical multisig as the load-bearing authority for mint authorisation. Legacy multisig remains valid for inbound migration messages from BSC during a 90-day sunset.

Concurrently the ZOO L2 chain ships a binary release that includes RINGTAIL verifier and ML-DSA verifier precompiles, so application contracts can begin signing with PQ keys.

7.3 Phase 2 — DAO migration (2025-Q3 to 2025-Q4)

The DAO ceremony per §6 runs in this window. Treasury custody, contract-upgrade authority, and proposal-signing keys all rotate to threshold PQ. The dual-key wallet update ships to all eligible voters.

Per-LLM chain governance keys rotate per the same template, in parallel with the global DAO rotation. Each per-LLM chain runs its own DKG; rotation order is by chain age (older chains first, so the highest-value chains are migrated first).

7.4 Phase 3 — zLLM and NFT Liquidity migration (2025-Q4)

- Experience-library accumulator commitments re-signed under threshold ML-DSA (§3).
- Inference-receipt schema cuts over to the 3.0 form.
- Outstanding NFT loans walked, counterparties co-sign 3.0 agreements (§5).
- Liquidation oracle re-keys to threshold ML-DSA.

7.5 Phase 4 — spec freeze and full-PQ activation (2025-10-31)

This paper, the supporting ZIPs (zLLM PQ commitments, NFT Liquidity PQ, Bridge PQ, DAO PQ migration), and the precompile-set release freeze on the founder’s anniversary date 2025-10-31. After this date only spec-clarification PRs are accepted.

The full-PQ enforcement cutover — after which legacy-only signatures fail validation — is staged through Q4 in line with the sunset windows defined per subsystem:

Subsystem	Sunset end	Behaviour after
Bridge committee	90 days post Phase 1	PQ-only mint authorisation
DAO governance	90 days post Phase 2	PQ-only ballots accepted
zLLM commitments	90 days post Phase 3	PQ-only experience admissions
NFT loans	90 days post Phase 3	PQ-only new loans; legacy flagged

7.6 Phase 5 — Quasar 3.0 inheritance (2025 cutover)

LUX QUASAR 3.0 ships its triple-cert lane in 2025 [4]. ZOO L2 settlement contracts begin requiring QUASAR 3.0 cert references on every batch finalisation. Light clients adopt the triple-cert verifier (§2). At this point the entire ZOO stack is end-to-end post-quantum.

7.7 Phase 6 — handoff to 4.0 (2026-02-14)

ZOO 4.0 [11] graduates ZOO from L2 to sovereign L1 on February 14, 2026. The 4.0 chain inherits the entire 3.0 PQ stack unchanged; the 4.0 work is graduating execution to GPU-native, not re-doing PQ. This is intentional: PQ is a 2025 commitment, GPU-native is a 2026 commitment, and the two epochs do not overlap.

7.8 Out-of-scope for 3.0

The following remain out of scope for 3.0 and ship in 4.0 or later:

- L1 graduation (Quasar-Native App Stack [11]).
- GPU-native consensus and EVM execution [1], [11].
- Native securities trading on Zoo DEX (April 20, 2026).

7.9 Cross-paper relationships

ZOO 3.0 sits between two papers:

- Upstream: ZOO 2.0 [10] (October 2024 L2 re-launch). The PQ precompile suite that 3.0 promotes from application-level option to chain-wide default originates in 2.0.
- Downstream: ZOO 4.0 [11] (February 14, 2026 L1 graduation). 4.0 inherits 3.0’s PQ posture unchanged.

- Lateral: Zoo 2025 securities and DAO [13]. The holographic-consensus DAO mechanics referenced for vote weighting and FHE ballots are designed to compose with 3.0's PQ governance keys.

7.10 One-line summary

ZOO 3.0 is the year of full PQ: every signature, every commitment, every proof in the ZOO stack moves to post-quantum schemes. By the end of 2025 the chain is end-to-end harvest-now-decrypt-later resistant. October 31, 2025 (founder's anniversary).

References

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- [3] Lux Foundation. LP-017: Threshold Bridge Committee Key Schedule. Lux Improvement Proposal, 2025.
- [4] Lux Foundation. LP-020: QUASAR Consensus 3.0 (Triple Consensus: BLS + RINGTAIL + ML-DSA). Lux Improvement Proposal, 2025.
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- [10] Zoo Labs Foundation. Zoo EVM: A Post-Quantum Safe L2 Blockchain with AI-Native Precompiles on the LUX Network. Zoo Whitepaper, October 2024.
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